Library Management System

Abstract

The Library Management System is a simple Microsoft Access based database application designed to record and manage the day to day activities of a college library. It keeps track of the books held by the library, the members who borrow them, and every issue and return transaction, so that the librarian can work from a single screen instead of a set of registers.

The database consists of five related tables — Authors, Books, BookIssues, Members and Publishers — linked through one-to-many relationships so that the same author, publisher or member is never repeated inconsistently. A book is catalogued once and then issued many times, which keeps the issue history complete without duplicating the book details. Records are entered through a set of user friendly forms rather than by typing directly into the tables: the main menu opens the member form, the book form and the issue/return form, and each form validates and saves its own record.

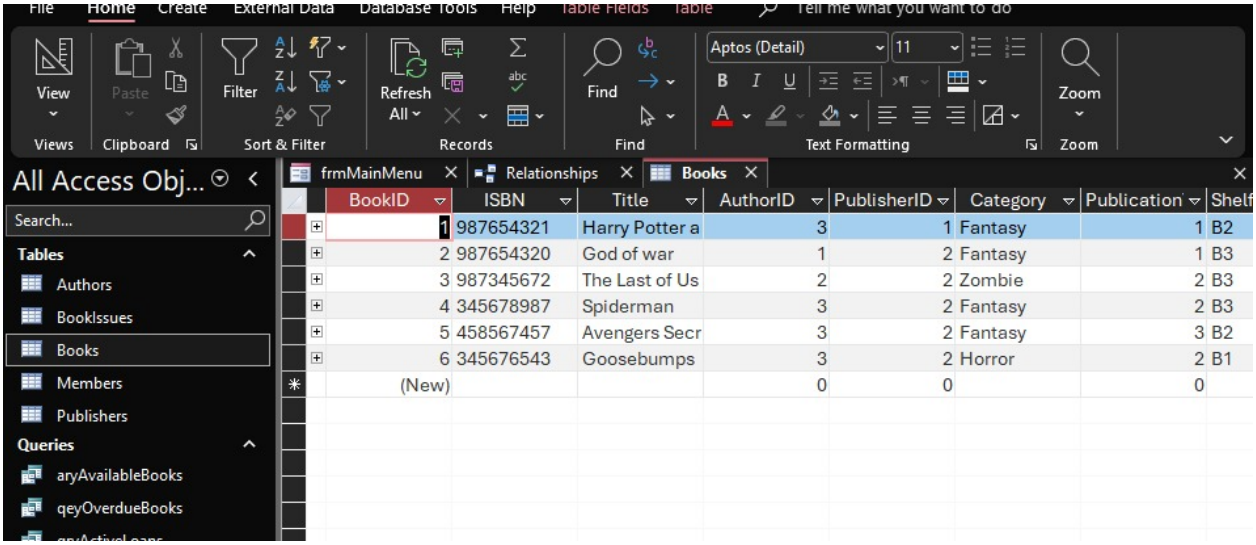
A query module is used to generate the information the librarian needs — the complete book catalogue, the list of books currently available, the active loans, the overdue books, and the borrowing history of each member. These queries feed a report that groups the overdue books by the month in which they fell due and lists the member's name and phone number beside each one, so that reminders can be sent quickly.

Overall, the project demonstrates the practical use of Microsoft Access for building a small scale information system that is efficient, easy to use, and effective for library management.

Screenshots

Tables

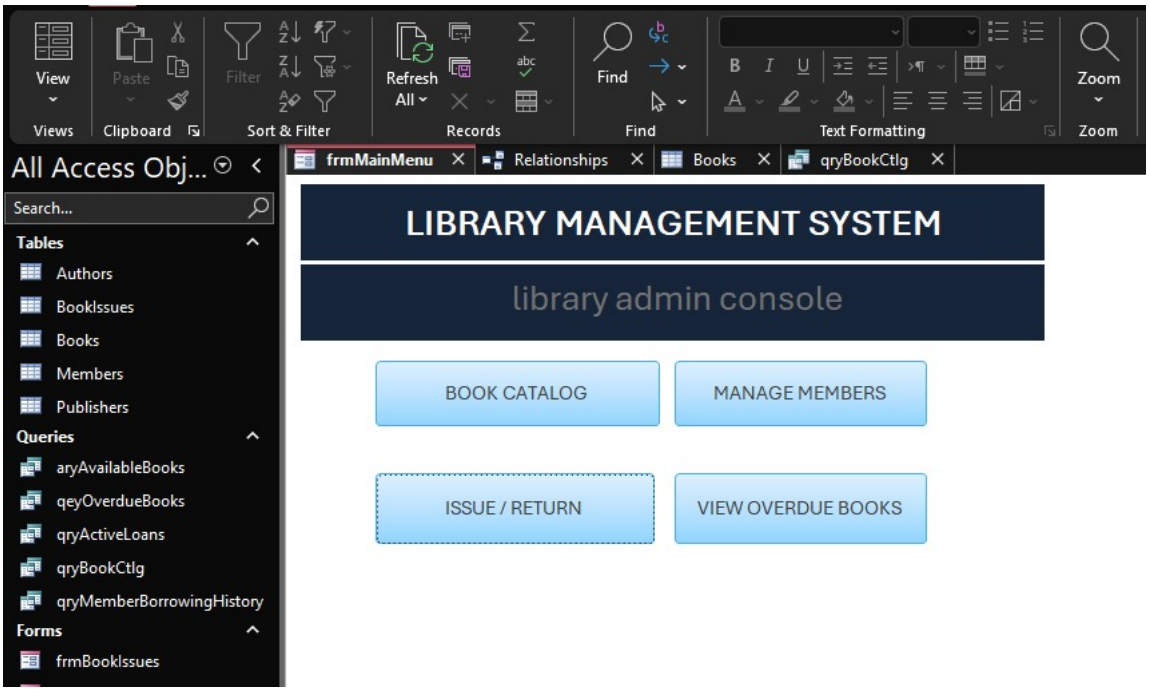
1. Books Table



Forms

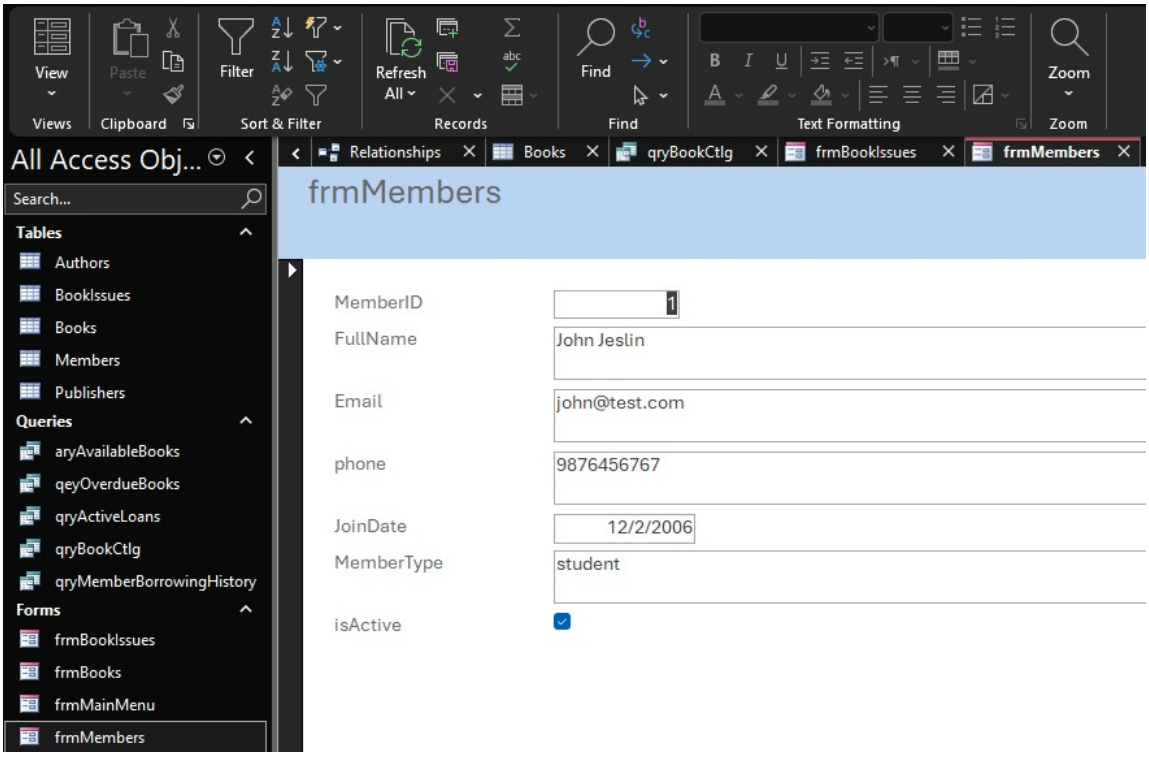
1. Main Menu Form

The main menu is the starting point of the application. It opens the book catalogue, the member form, the issue/return form and the overdue books report from a single window.



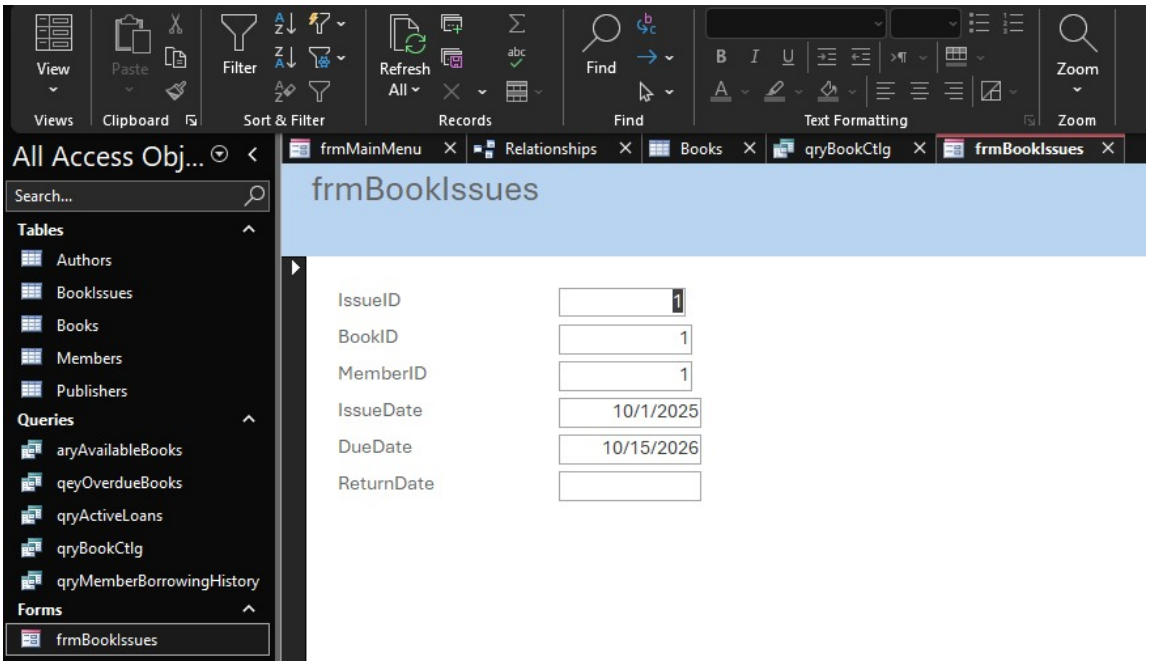
2. Members Form

The members form is used to add and edit library members, recording the name, email, phone number, join date and member type, and whether the membership is currently active.



3. Book Issues Form

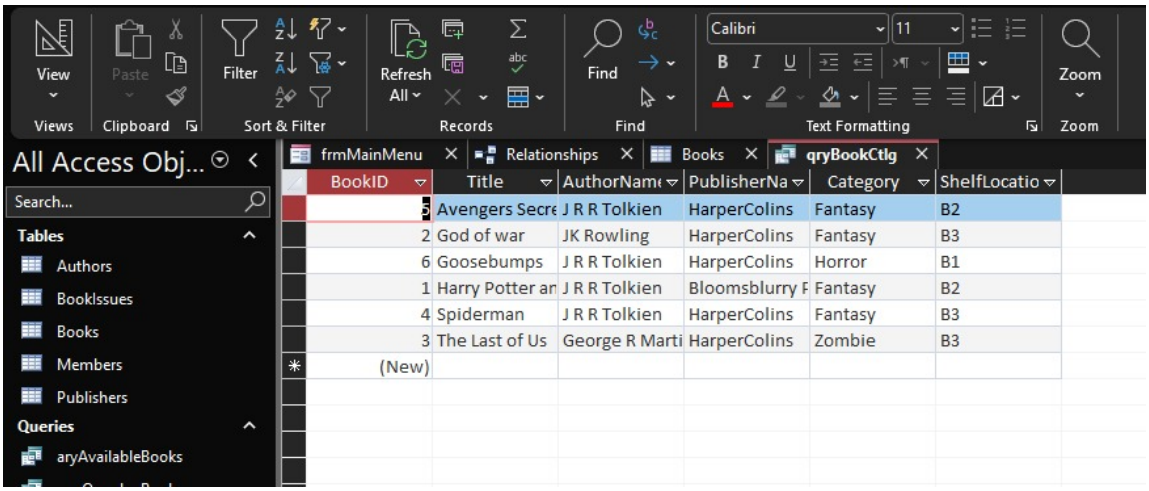
The book issues form records a loan against a book and a member. It stores the issue date and the due date, and the return date is filled in when the book comes back.



Queries

1. Book Catalogue Query

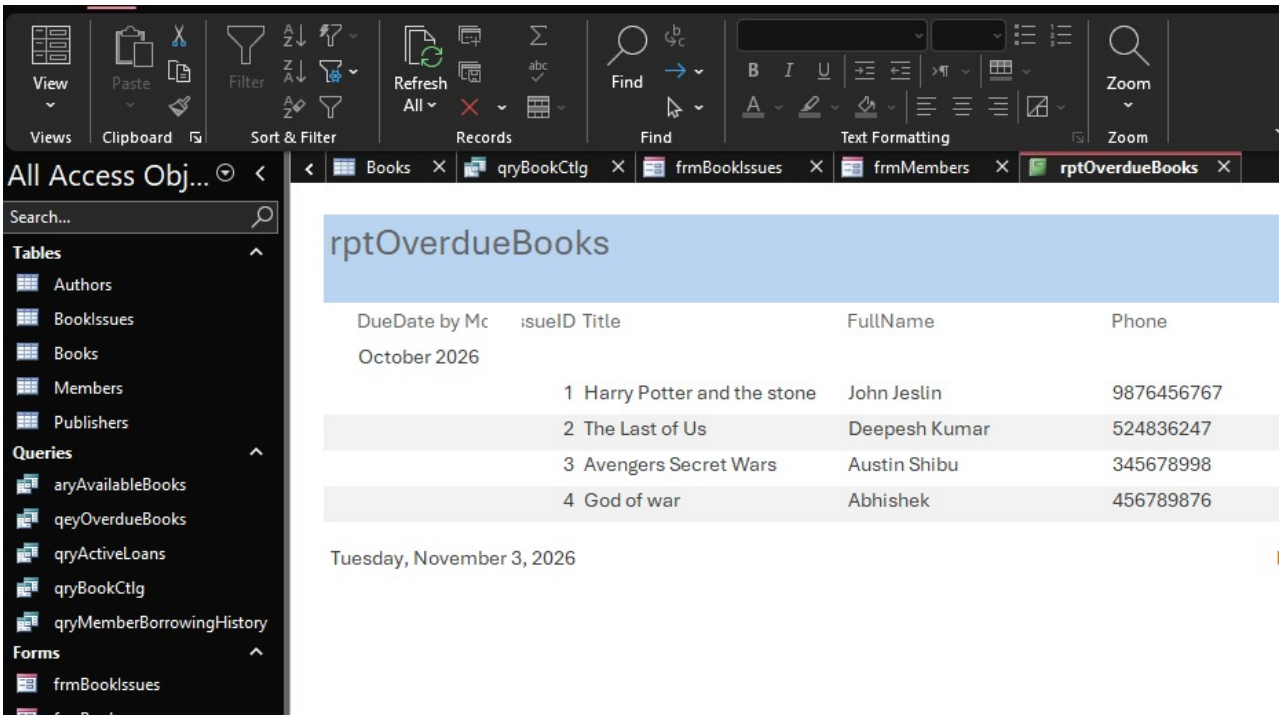
The book catalogue query lists every book with its author, publisher, category and shelf location, so that the shelf can be found without opening the individual tables.



Reports

1. Overdue Books Report

The overdue books report lists the books that were not returned by their due date, grouped by the month in which they fell due, with the member's name and phone number so that a reminder can be sent.



Relationships

The relationships window shows how the five tables are joined. Each author and each publisher is related to many books, each book and each member is related to many issues, and every relationship enforces referential integrity.

